

CONTACT

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Dr. Daniel Mesafint Belete

Assistant Professor - Computer Science
Debre Tabor University, Ethiopia

ABOUT ME

"Passionate Scholar and Advocate of Computer Science Education"

Transforming the landscape of computer science through education, research, and mentorship. Are you on a similar path? Let's connect and make an impact!

I boast a prosperous academic career holding a **Ph.D.**, **MSc**, and **BSc**, underpinned by sixteen fruitful years in academia and research. This journey has allowed me to publish extensively, with over eight articles in esteemed academic journals, contributing rigorously to the scientific community.

My real passion lies in mentoring; supervising MSc and Ph.D. candidates is more than a role; it's a commitment to empower and inspire. It's a platform where I can share my love for innovation and problem-solving, helping to sculpt the upcoming generation of computer science experts.

Let's inspire and drive impactful changes together!

EDUCATION

Computer Science (Ph.D.) **2018 - 2022**
Mangalore University
Mangalore, Karnataka, India.

Ph.D. Thesis on: Exploring Predictive Models for the Analysis of the Epidemics of HIV/AIDS.

Information Technology (M.Sc.) **2011 - 2012**
Adama University
Adama, Oromia, Ethiopia.

MSc. Thesis on: Mining to Identifying telltale patterns of child labor survey.

Information Technology (B.Sc.) **2005 - 2007**
Mekelle University
Mekelle, Tigray, Ethiopia.

BSc. Project on: Web site development for St. Frumnutious Aba Selama Kessate Birhan Theological College.

SKILLS

- Academic Mentorship
- Innovation Management
- Educational Leadership
- Curriculum Development
- Effective Communications
- Collaborative Teamwork
- Creative Problem-Solving
- Personal Accountability
- Business Acumen
- Adaptability
- Research Skill
- Teaching

PROJECTS

Grant projects I worked on:

- A Multilingual Conversational AI in Healthcare with Chatbots and Virtual Assistants.
- AI-Driven Diagnostic Tool for Early Detection and recommendation of Diseases in Rural Ethiopia.
- AI-Driven Malaria Detection - Development and Validation of a Deep Learning-Based Diagnostic Tool.
- Facial Emotion Recognition for Spotify Music Recommendation System.

WORK EXPERIENCE

Research, Community Service and Technology Transfer & University-Industry Linkage Coordinator — **Jan, 2023 - Present**
Gafat Institute of Technology
Debre Tabor University
Debre Tabor, Amhara, Ethiopia

- Responsible for CED and report to the Vice President for research and technology transfer.
- Collaborate with the Vice President on research and technology transfer.
- Works in coordination with CAD, CCD and business and development.
- Responsible for research, community services and technology transfer activities.
- Identify and implement community engaging and problem-solving activities.
- Monitor the research results of the institution for publication.
- Directs projects; strives for consistency and clarity.
- Plans field stays for students; monitors and executes.
- Verifies according to the guidelines before conducting various research questions.
- Provides up-to-date reports to CED and Vice President for research and technology transfer.
- Performs additional assignments from CED.

Assistant Professor — **Jan, 2015 - Present**
Department of Computer Science & IT
Debre Tabor University
Debre Tabor, Amhara, Ethiopia

- Facilitated and moderated computer science and information technology related course discussions in alignment with the curriculum.
- Designed and revised syllabus, curriculum, instruction content, seminars/tutorials and other learning materials as needed.
- Provided individualized counseling and supervision to students in laboratory work and course-related issues.
- Maintained expertise in advancements in computing, technology, programming, and syntax.
- Participated in department administration and faculty meetings, wrote reports, and managed research budgets.
- Conducted research and wrote research proposals, papers, and other publications, and supervised MSc students and research staff.
- Contributed to departmental planning and operations, as well as the development and implementation of university policies and laws.

TAUGHT

Here is the subject I taught but not limited to:

- Machine Learning.
- Deep Learning.
- Big Data.
- Internet of Things (IoT).
- Distributed Database Systems.
- Wireless Communication and Mobile Computing.
- Software Project Management and Information Technology.
- Research Methodology and Scientific Writing.
- Advanced Computer Networks.
- Advanced Requirement Engineering.
- Advanced Concepts in Data Mining.
- Service Oriented Architecture.
- Cloud Computing.
- Semantic Web Technology.
- Operating Systems (OS).
- System Analysis and Design.
- Natural Language Processing.
- Web Programming.

Information Technology Instructor/Trainer — **Sep 2009 - Aug 2015**
Department of Information Technology
Debre Poly Technique College
Debre Tabor, Amhara, Ethiopia

- Identifying technical training needs and requirements for individuals and organizations.
- Developing instructional training materials such as handbooks, visual aids, online tutorials, demonstration models, and reference documentation.
- Designing, coordinating, and conducting ICT training programs, delivered through individual and group instruction and workshops.
- Liaising with external training providers to arrange delivery of specific training programs.
- Providing on-site support for professors and faculty in technology-assisted lectures and presentations.
- Monitoring and evaluating training quality and effectiveness and making modifications to training objectives, methods, and course deliverable.
- Assisting supervisors with projects such as developing interactive Amharic learning software and organizing computer-based curriculum materials for classes.

Information Technology Instructor/Trainer — **Sep 2008 - Aug 2009**
Department of Information Technology
Almaz Boum Technical & Vocational College
Alem Ketema, Amhara, Ethiopia

- Designed and developed instructional training materials, including user guides, quick reference guides, videos, and online learning.
- Contributed to the development and delivery of the HR systems transformation training program through analysis of training needs, creating training plans, and tailoring courses.
- Evaluated training effectiveness and made modifications to approach and content as necessary.
- Conducted virtual training sessions and engaged with stakeholders to understand their learning needs.

Part-Time Lecturer — **Sep 2010 - Aug 2012**
Department of Information Technology
Fekede-Egzi College
Debre Tabor, Amhara, Ethiopia

- Preparing and delivering lectures, tutorials, workshops, and seminars.
- Developing curricula and course material that can be used across a number of platforms.
- Collaborating with other academics and lecturers to improve teaching methods and expand knowledge base.
- Setting and grading assignments, tests, and exams.
- Conducting research, and writing papers, proposals, journal articles, and books.
- Attending and participating in meetings, conferences, and other events in and outside of the institution.
- Participating in training opportunities and initiatives at the institution.
- Providing support to students and other colleagues.

INTERESTS

Area of research interest:

- Adversarial Network
- Chatbot
- Conversational Agents
- Machine Learning
- Deep Leaning
- Big Data
- Feature Extraction
- Data Mining
- Data Warehouse
- Artificial Intelligent
- Natural Language
- Internet of Things (IoT)

PEER REVIEW

As a reviewer for various esteemed journals:

- I have critically evaluated numerous research articles, ensuring the dissemination of high-quality scholarly work on:
  - Transactions on Knowledge Discovery from Data (ACM).
  - BMJ Public Health.
  - Journal of Big Data.
  - Informatics in Medicine Unlocked.
  - MethodsX.
  - IEEE Access.
  - Environmental Monitoring and Assessment.
  - International Journal of Pharmaceutical and Biomedical Science.
  - Journal of Medical Research and Health Sciences.
  - International Journal of Electrical and Electronics Research (IJEER).
  - East Africa Journal of Social and Applied Sciences(EAJSAS).
  - African Development Review.

Deliver Tutor for Distance Students — **Sep 2010 - Aug 2011**
Department of Information Technology
St. Mary's College
Debre Tabor, Amhara, Ethiopia

- Tutored students in all subject areas in which they required assistance.
- Used a variety of teaching methods depending on individual learning style.
- Assisted with test preparation homework help study skills and improved note-taking.
- Provided instruction in an asynchronous online learning format.
- Conducted needs assessments and designed individualized tutoring services according to student needs.
- provide advice on different levels of access to learning materials according to the needs of individual participants.

Registrar Evaluation Director — **2009 - 2010**
Debretabor Polly Technique College
Debre Tabor, Ethiopia

As Registrar Evaluation Officer, key role to Organizing and administering student records, Overseeing the student admissions and graduation process, Ensuring records are updated with new grades, attendance, finances, etc., Training staff at the registrar's office to use software related to records administration, Performing clerical tasks, such as printing academic transcripts for students, Keeping student information confidential and secure, and Participating in student service committees and initiatives.

Business Department Head — **2010 - 2011**
Debretabor Polly Technique College
Debre Tabor, Ethiopia

As Business Department Head, key role to work proactively in the application and Academic Infrastructure to work in the Department, Contribute to the development in approaches to learning and teaching within the department and serve on the college's Learning and Teaching staffs, Lead and oversee the Department's Open Day activity, ensuring adequate staff coverage for all of the programmes in the Department, Manage and provide regular updates on Workload Allocation within the Department, maintaining accurate Workload Allocation Records, Ensure that all staff within the Department have accessed up to date training in areas relevant to their role such as Mental Health and Health and Safety, and Responsible for all aspects of health and safety within her/his department.

Information Technology Department Head — **2013 - 2014**
Debretabor Polly Technique College
Debre Tabor, Ethiopia

As ICT Department Head, key role to have overall responsibility for all teaching and learning within the department, Ensure that all members of the department are performing as effectively and efficiently as possible and ensure to the best of her/his ability, that the department is adequately resourced to fulfil its function within the college, Keep abreast of national strategy and developments impacting on her/his subject area/s and ensure information is communicated to the department, and the department is responsive to such developments, and Contribute as appropriate to whole-school initiatives such as induction days and shall encourage extra-curricular activities and educational visits within the college guidelines.

SUPERVISION

Mentoring and supervising MSc and Ph.D. scholars on:

- Assessing the Challenges of Machine Learning Practitioners' towards a Robust and Trustworthy Machine Learning System.
- Developing Conversational Agent for Medical Support and Diagnosis.
- Multi Class Hate Speech Detection and Classification from Social Media Posts and Comments in Amharic Text.
- Identification of Classroom Sheet Exam Cheating Trials in Exam Video Data Using a Deep Learning Approach.
- A Deep Learning Model for Amharic Audio Based Virtual Assistant.
- Auto Answering Model Design for Diabetic Disease in Amharic Language Using Deep Learning.
- Automatic Amharic Text Categorization From Social Media Data Using Deep Learning.
- Prediction Model For Maternal Delivery Mode Using Machine Learning.
- Emotion Recognition Based On Facial Expression Analysis.
- Medical-Insurance Fraud Detection.
- Designing Automatic Question Answering Model For Health Diagnosis Using Deep Learning Approaches.
- Machine Learning Techniques for Customer Churn Prediction in Commercial Bank of Ethiopia (CBE).
- Designing a Rule-Based Expert System Model for the Diagnosis and Treatment of HIV/AIDS.
- Hybrid Fake News Detection on Social Media.
- Exploring Amharic Language Models for Machine Translation and Sentiment Analysis.
- Enhanced Price Prediction of Seasonal Agricultural Products Using Deep Learning.
- Real-Time Car Fuel Consumption Monitoring System Integrated with IoT.

PUBLICATIONS

Paper Title: **Grid search in hyperparameter optimization of machine learning models for prediction of HIV/AIDS test results** — **SCOPUS Indexed**
Journal Name: **International Journal of Computers and Applications**

- **ISSN:** N/A, **Vol** 44, **Issue** 9, **pp** 875-886, Year 2021
- **Publisher:** Taylor & Francis
- **Available on:** https://doi.org/10.1080/1206212X.2021.1974663

Paper Title: **Deep Learning Models for Predicting HIV Test Status using DHS data** — **SCOPUS Indexed**
Journal Name: **Future Opportunities and Tools for Emerging Challenges for HIV/AIDS Control**

- **ISSN:** 2631-6188, **pp** 1-21, Year 2023
- **Publisher:** IntechOpen
- **DOI:** 10.5772/intechopen.104224
- **Available on:** https://www.intechopen.com/chapters/81573

Paper Title: **Performance Evaluation of Classification Models** — **SCOPUS Indexed**
Journal Name: **Data Management, Analytics and Innovation**

- **ISSN:** 978-981-16-2934-1, **Vol** 70, **pp** 109–125, Year 2021
- **Publisher:** Springer
- **DOI:** 10.1007/978-981-16-2934-1_7
- **Available on:** https://link.springer.com/chapter/10.1007/978-981-16-2934-1_7

Paper Title: **Wrapper Based Feature Selection Techniques On EDHS-HIV/AIDS Dataset** — **SCOPUS Indexed**
Journal Name: **European Journal of Molecular & Clinical Medicine**

- **ISSN:** 2515-8260, **Vol** 07, **Issue** 08, **pp** 2642-2657, Year 2020
- **Publisher:** eJournalPlus
- **Available on:** https://ejmcm.com/article_4776.html

Paper Title: **A Comparative Study of Filter and Wrapper Methods on EDHS – HIV/AIDS Dataset** — **SCI Indexed**
Journal Name: **Proceedings of the Third International Conference on Smart Systems and Inventive Technology (ICSSIT 2020)**

- **ISSN:** 978-1-7281-5821-1, **pp** 1264-1271, Year 2020
- **Publisher:** IEEE
- **DOI:** 10.1109/ICSSIT48917.2020.9214212
- **Available on:** https://ieeexplore.ieee.org/document/9214212

- Ensemble Models for Enhanced Stock Market Prediction.
- Predicting Effective Metrics of It Projects in Software Enterprise Using Machine Learning Approach.
- Exploring Deep Learning Model for Fraud Detection in Motor Insurance.
- Prediction of Mobile Network Congestion Using Deep Learning in Case of Ethio-Telecom.
- Prediction of Liver Disease Using Machine Learning Techniques.
- Ethiopian Currency Classification Using Neural Network Approach.
- Development of an Efficient and Adaptable Fine-Grained Car Image Classification Model Using Multimodal Data and Advanced Deep Learning Techniques.
- Sentiment Analysis for Himtagna Language: A Deep Learning Approach for Low-Resource Language Processing.
- Part of Speech (PoS) Tagging for the Harari Language using Advanced Deep Learning Approaches.
- Vehicle Reliability Prediction Models Development And Failure Modes and Effects Analysis.

SCI Indexed

Paper Title: **Classification of Binary Class HIV/AIDS Test Results Using Ensemble Learning Models**
Journal Name: **International Conference on Decision Aid Sciences and Application (DASA)**

- **ISSN:** 978-1-7281-9677-0, **pp** 1240-1247, Year 2020
- **Publisher:** IEEE
- **DOI:** 10.1109/DASA51403.2020.9317082
- **Available on:** https://ieeexplore.ieee.org/document/9317082

UGC Indexed

Paper Title: **Feature Selection Methods for Prediction of the Individual's Status of HIV/AIDS from EDHS dataset - A Filter Approach**
Journal Name: **International Journal of Research in Engineering Advanced Technology (IJREAT)**

- **ISSN:** 2320 – 8791, **Vol** 07, **Issue** 03, **pp** 29-36, Year 2019
- **Publisher:** Pioneer research & development group
- **Available on:** http://www.ijreat.org/Issue39.html

UGC Indexed

Paper Title: **Predictive Models of HIV/AIDS Epidemic Status using Data Mining Techniques: A Review**
Journal Name: **Journal of Emerging Technologies and Innovative Research(JETIR)**

- **ISSN:** 2349-5162, **Vol** 06, **Issue** 05, **pp** 10-16, Year 2019
- **Publisher:** IJ Publication
- **Available on:** https://www.jetir.org/view?paper=JETIRCU06003

CONFERENCES & WORKSHOPS

November 2020

IEEE, International Conference on Decision Aid Sciences and Applications (DASA'20)
University of Bahrain, Bahrain

June 2020

Online International Conference on "Recent Trends in Computer Science and Information Technology", (ICRCSIT-2020)
St. Martin's Engineering College, Dhulapally, Secunderabad, India

July 2020

Third International Conference on Emerging Trends in Science & Technologies for Engineering Systems, (ICETSE-2020)
S J C institute of Technology, Chickballapur, Karnataka, India

COMPETENCY

Competency of Excellence: Level IV

Information Technology Assistance

For having demonstrated competence against the Ethiopian Occupational Standard for Information Technology Assistance with Level-IV, issued the authority of the Ministry of Education and the National TVET Qualification Framework.

Competency of Excellence: Level III

Information Technology System Support

For having demonstrated competence against the Ethiopian Occupational Standard for Information Technology System Support with Level-III, issued the authority of the Ministry of Education and the National TVET Qualification Framework.

Competency of Excellence: Level IV

Database Administration

For having demonstrated competence against the Ethiopian Occupational Standard for Database Administration with Level-IV, issued the authority of the Ministry of Education and the National TVET Qualification Framework.

CISCO: CCNA I

Networking for Home and Small Businesses

Completing the CCNA Discovery: Networking for Home and Small Businesses course as part of the Cisco Networking Academy.

CISCO: CCNA II

Working at a Small to Medium Business or ISP

Completing the CCNA Discovery: Working at a Small-to-Medium Business or ISP course as part of the Cisco Networking Academy.

IEEE, Third International Conference on Smart Systems & Inventive Technologies (ICSSIT 2020) — August 2020
Francis Xavier Engineering College, Tirunelveli, India.

Springer, Fifth International Conference on Data Management, Analytics & Innovations, (ICDMAI-2021) — January 2021
Kolkata, India

CERTIFICATIONS

Modeling fundamentals and Python for numerical simulation and AI for Mechanical Engineering System Research — January 2021
NITK College
Mangalore, India

has successfully completed one week training on "Modeling fundamentals and Python for numerical simulation and AI for Mechanical Engineering System Research", organised by Department of Computer Science and Information Technology, conducted from 18-23/2021.

Big Data Processing using Apache Spark — August 2020
JNN College of Engineering
Shivamogga, Karnataka, India

has successfully attended Webinar on " Big Data Processing using Apache Spark", organised by Department of Information Science and Engineering, JNN College of Engineering, Shivamogga, Karnataka, in association with Computer Society of India(CSI) conducted on 08/08/2020.

E-learning using LMS tool — September 2020
Sri-Krishina Institute of Technology
Bangalore, India

has successfully attended Webinar on "E-learning using LMS tool", organised by Department of Information Science and Engineering, SKIT, Bangalore on 03/09/2020

Digital System Design with ASM Charts — May 2020
Holy Marry Institution of Technology
Hyderabad, India

has successful participation in Online Quiz on "Digital System Design with ASM Charts", organised by Department of Electronics and Communication Engineering conducted on 5/30/2020, with a passing score of 95%

Operating Systems Skills — June 2020
JAI Shriram Engineering College
Tirupur, India

has successfully completed on "Operating Systems Skills" conducted by Department of Computer Science and Engineering , Jai Shriram Engineering college on 7/6/2020

CISCO: CCNA III

Introducing Routing and Switching in the Enterprise

Completing the CCNA Discovery: Introducing Routing and Switching in the Enterprise course as part of the Cisco Networking Academy.

CISCO: CCNA IV

Designing and Supporting Computer Networks

Completing the CCNA Discovery: Designing and Supporting Computer Networks course as part of the Cisco Networking Academy.

Digital System Design

Online Test on Digital System Design with ASM Charts

A certificate issued by AICTE, accredited by NAAC "A" Grade College to prove the skills of Digital System Design with Score of 95%.

Operating Systems

Test your Operating Systems Skills

A certificate issued by approved by AICTE, affiliated to Anna university, accredited by NAAC to prove the skills of Operating System with Score of 100%.

Network Simulations Using NS-3 - A Research Perspective — **June 2020**
SRI College of Engineering
Bangalore, India

For active participation in National Level Online Faculty Development Program on "Network Simulations Using NS-3 - A Research Perspective" and cleared assessment with "A" Grade held on 6/6/2020 organized by Department of Information Science & Engineering.

Artificial Intelligence and Image Processing Using MATLAB — **July 2020**
EGS Pillay Engineering College
India

has successfully completed the online 2 days Webinar Course in Artificial Intelligence and Image Processing Using MATLAB from 09/07/2020 to 10/07/2020

Artificial Intelligence-Present State and Future Perspectives — **September 2020**
Assumption College
Kerala, India

has successfully participated in the International Webinar "Artificial Intelligence-Present State and Future Perspectives" conducted from 24-26, Sep 2020.

Artificial Intelligence-Present State and Future Perspectives — **June 2020**
JNN College of Engineering
Shivamogga, India

has successfully attended Webinar on "5G and Beyond" , organised by Department of Information Science and Engineering, JNN College of Engineering, Shivamogga, Karnataka, in association with Computer Society of India(CSI) conducted on 28/07/2020.

PHP & MySQL — **May 2020**
JNTUH College of Engineering
Shivamogga, India

has successfully completed one week training on "PHP & MySQL", organised by Department of Computer Science and Engineering, JNTUH College of Engineering, Shivamogga, Karnataka, in association with Computer Society of India(CSI) conducted from 18-23-2020.

Artificial Intelligent and Machine Learning — **February 2020**
NITK College
Mangalore, India

has successfully completed one week training on "Artificial Intelligent and Machine Learning", organised by Department of Computer Science and Information Technology conducted from 10-15/2020.

ACTIVITIES

Actively engaged in the various committees on:

- Research Guidelines Preparation Team for Debre Tabor University Research Center.
- Technology Transfer Committee's Team Leader.
- Academic Committees of Gafat Institution of Technology.
- National Qualification Competency Assessors for IT competencies in Center of Competency (COC) agency.
- Discipline Committees Member in University.
- Regional TVET Curriculum Designer.
- BOE Member in Computer Science/IT at undergraduate level.
- Academic Council Member in Debre Tabor University.
- Curriculum Designing Committee Members for MSc program in Debre Tabor University.

Internet of Things, IBM — **February 2020**
Mangalore University
Mangalore, India

has successfully completed one week training on "Internet of Things, IBM", organised by Department of Electrical and Electronics conducted from 03-08/2020.

CCNA Discovery: All Packages, (CISCO) — **Feb - May 2014**
Adama University
Adama, Ethiopia

has successfully completed all 4 courses on "CCNA Discovery: All Packages", organised by Department of Computer Science conducted from Feb - May, 2014.

Training Methodology for B-LEVEL TVET Teachers — **Apr - May 2010**
Ethio-China Poly Technique College
Addis Ababa, Ethiopia

has successfully completed one month training on "Training Methodology for B-LEVEL TVET Teachers", organised by Ethio-China Poly Technique College conducted from Apr, 12 to May, 04/2010.

Network Fundamentals, Network Administrations and Computer Maintenance & Troubleshooting — **Jul - Aug 2009**
Mekelle Institution of Technology (MIT)
Mekelle, Ethiopia

has successfully completed one month training on "Network Fundamentals, Network Administrations and Computer Maintenance & Troubleshooting", organised by MIT conducted from Jul, 06 to Aug, 07/2009.

Information Technology TVET Curriculum Design — **Nov - Dec 2008**
Gondar TVET College
Gondar, Ethiopia

has successfully participated one month workshop on curriculum development on "Information Technology Occupation", organised by Amhara region TVET agency conducted from Nov, 10 to Dec, 03/2008.

14th March 2025

Dr. Daniel Mesafint

REFERENCES

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Dr. Ibrahim Gashaw Kassa

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